

Reaction-Diffusion Simulation of Energy-Constrained Intra-Body Nanosensor Networks

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Abstract—Intra-body nanosensor networks represent a promising paradigm for continuous health monitoring and targeted drug delivery, but their practical deployment is fundamentally constrained by energy. This paper extends reaction-diffusion simulation for intra-body nanosensor networks by representing nanosensor energy as a per-unit state variable that is updated during simulation and used to gate detection and local communication reactions: a nanosensor without sufficient energy cannot detect biomarkers or propagate gossip, and its behaviour changes accordingly in real time. We implement this model in AcCoRD (Actor-based Communication via Reaction-Diffusion), a high-fidelity hybrid micro-/mesoscopic simulator, through per-unit energy reservoirs, molecule ownership tracking, energy-gated reactions, and a biomarker identity set (BmIDSet) mechanism for information deduplication. The extended simulator is tested on a five-region branching arteriole scenario with 56 biomarker molecules and 100 nanosensors organised in four clusters. Simulation results confirm that nanosensors involved in more detection and gossip events deplete their energy faster, and that the BmIDSet mechanism faithfully tracks the spread of biomarker information through the network.

Index Terms—Molecular communication, nanosensor networks, energy modelling, AcCoRD, reaction-diffusion simulation, Internet of BioNanoThings, biomarker detection, gossip protocol.

I. INTRODUCTION

THE Internet of BioNanoThings (IoBNT) envisions a body-area network of sub-micron devices embedded in biological fluids that cooperate to sense, compute, and communicate health-relevant information [1]. Within this vision, nanosensors are the fundamental building blocks: tiny nodes that drift or adhere in blood or tissue, detect specific molecules (biomarkers) circulating in the fluid, and relay detection events to neighbouring nodes or to an external gateway.

A central challenge in realising such networks is energy. Unlike conventional wireless nodes that can harvest radio-frequency power or carry a battery, intra-body nanosensors must subsist on energy stored in a nanoscale reservoir (e.g., a piezoelectric nanogenerator or a biochemical fuel cell [2]). The reservoir replenishes slowly and is consumed by every computational and communication operation. If a nanosensor runs out of energy before it has completed its sensing task, information is lost and the system silently degrades. Modelling this phenomenon accurately requires a simulation environment that couples the physical diffusion of molecules, the stochastic chemistry of biomarker-nanosensor binding, the molecular signalling between nanosensors (gossip), and the energy bookkeeping of each individual node.

AcCoRD [3] is a state-of-the-art open-source reaction-diffusion (RD) simulator for intra-body MC, offering particle-level accuracy for diffusion and chemical reactions. However, it does not include a per-node energy model coupled to reaction events: energy is at best an offline post-processing concern, not an in-loop constraint that determines whether a node can react. If energy is computed only after the run, a depleted nanosensor can still participate in later simulated reactions. The question addressed in this paper is therefore how to make nanosensor energy an internal simulation state that can affect subsequent reaction dynamics, rather than a cost estimated after the simulation has already run.

The main contributions of this paper are:

- 1) We introduce a nanosensor energy model for reaction-diffusion simulation, where each nanosensor maintains an individual energy reservoir updated by passive drain and event-based costs for detection and gossip communication.
- 2) We integrate nanosensor energy state through molecule ownership and energy-gated reactions, allowing depleted nanosensors to stop participating in subsequent detection and gossip events.
- 3) We add biomarker identity tracking through a *biomarker identity set* (BmIDSet), enabling the simulator to distinguish unique biomarker information from repeated propagation and to reconstruct information spread across the nanosensor network.
- 4) We demonstrate the extension in an arteriole flow scenario, showing how local reaction activity leads to different energy trajectories, depletion events, and information propagation patterns across nanosensor clusters.

The remainder of the paper is structured as follows. Section II introduces the background concepts. Section III reviews related work. Section IV details the implementation. Section V presents and discusses simulation results. Section VI concludes the paper.

II. BACKGROUND

A. Diffusion-Based Molecular Communication

In diffusion-based MC, information is encoded in the type, concentration, or timing of *information molecules* (IMs) released by a transmitter into a fluid medium. IMs propagate by Brownian motion (a thermally-driven random walk that requires no additional energy from the emitter) and are detected by a receiver nanosensor upon physical contact with its

surface. In the RD microscopic regime this contact condition is implemented via a *binding radius* $r_b = r_{\text{BM}} + r_{\text{NM}}$, the sum of the radii of the two reacting species: a reaction fires when the centre-to-centre distance falls below r_b , which is equivalent to the two molecules touching. The binding radius therefore models the physical size of the molecule (nanomachine or biomarker), not an abstract interaction range. Detection is modelled as a bimolecular reaction:



where BM denotes a biomarker molecule and NM denotes a molecule representing the receptor state of the nanosensor. The reaction rate k is set high enough (approaching infinity) to model perfect absorption upon contact. In this work, reaction events are not only chemical transitions: they are also the points where nanosensor state changes and energy costs are applied.

After detection, nanosensors may wish to alert their neighbours. This is modelled as a *gossip* reaction, where an informed nanosensor converts a neighbouring naive nanosensor to an informed state upon close contact:



where k_g is the gossip binding rate, set high enough to model instantaneous information transfer upon contact. This reaction propagates awareness of a detection event through the network without any electromagnetic infrastructure. Note that the gossip reaction is an abstraction of local information exchange between nanosensors: the molecular collision models proximity-triggered communication, not a specific biochemical signalling pathway.

B. AcCoRD Simulator

Reaction-diffusion (RD) simulation is the computational paradigm that jointly models the random motion of individual molecules and the chemical reactions they undergo upon encounter. At the microscopic scale, molecule positions are propagated by Brownian dynamics and reactions fire stochastically when two reactants come within a binding radius; at the mesoscopic scale, well-mixed subvolumes exchange molecules according to diffusion rates and reaction propensities. This two-regime approach balances spatial accuracy against computational cost, and is particularly well suited to intra-body MC where biomarker densities vary sharply across vessel segments.

AcCoRD [3] is an open-source hybrid micro-/mesoscopic simulator for 3-D reaction-diffusion systems. Its key abstractions are:

- **Regions:** 3-D volumes (rectangular boxes or spheres) simulated either in the fine-grained *microscopic* regime (continuous Brownian motion, time step Δt) or the coarser *mesoscopic* regime (well-mixed subvolumes, discrete-time stochastic reactions). Regions can be nested and connected.
- **Actors:** logical entities that either release molecules (*active* actors) or observe them (*passive* actors). Multiple actors can overlap a region.

- **Chemical reactions:** specified globally or per-region, supporting zeroth-, first-, and second-order kinetics. Bimolecular reactions in the microscopic regime use a binding radius r_b .

The simulator reads a JSON configuration file and writes observations and a machine-readable summary to output text files. All stochastic processes are seeded with a user-controllable random seed, enabling reproducible experiments.

Energy constraints. Kuran *et al.* [4] derived a model for MC emission based on the cellular exocytosis pathway. Emission involves four stages: (1) molecular synthesis (dominant cost, proportional to peptide length); (2) vesicle formation; (3) vesicle transport along microtubules; and (4) membrane fusion. Using 1 ATP $\approx$ 83 zJ [5], the total emission energy for n insulin molecules is:

$$E_T = nE_S + \left\lceil \frac{n}{c_v} \right\rceil (E_V + E_C + E_E), \quad (3)$$

where $E_S \approx 10$ aJ per molecule is the synthesis cost, E_V the vesicle formation energy, E_C the vesicle transport cost along microtubules, E_E the membrane fusion energy, and $c_v = 1540$ molecules per vesicle. The vesicle overhead $E_V + E_C + E_E \approx 66$ aJ per vesicle is negligible compared to the synthesis cost. This makes MC vastly more energy-efficient than electromagnetic alternatives at intravascular scales.

Existing simulation frameworks for intra-body molecular communication do not, to our knowledge, provide comprehensive in-loop tracking of energy consumption across the full range of nanosensor events: passive drain, biomarker detection, and inter-nanosensor gossip. The AcCoRD energy extension addresses this gap by coupling energy state directly to reaction dynamics, with configurable cost parameters that can be populated from any biophysical model as such data become available.

III. RELATED WORK

MC simulators. Several simulators have been proposed for nanonetwork research. N3Sim [6] is a microscopic MC simulator for diffusion-based communication. BloodVoyagerS [7] is an ns-3 module focused on nanobot movement in the human vascular system. MEHLISSA 2.0 [8] is a recently redesigned full-body MC simulator for the human circulatory system; by removing its ns-3 dependency it achieves over $2\times$ runtime reduction, enabling large-scale particle simulations, though it does not model per-nanosensor energy state. AcCoRD [3] distinguishes itself by combining: (i) particle-level microscopic simulation for accuracy in regions of interest; (ii) mesoscopic compartment-based simulation for computational efficiency elsewhere; and (iii) a flexible actor abstraction for modulation schemes including burst, concentration shift keying (CSK), and point sources.

Energy models. The energy model of Kuran *et al.* [4] remains the reference for MC emission costs. More recent works have analysed energy efficiency in multi-hop mobile MC systems [9] and derived energy-delay trade-offs in advection-diffusion body area networks [10]. Energy harvesting through piezoelectric effects has also been studied [2]. These works, however, treat energy analytically or as a post-processing step;

they do not couple energy state to reaction dynamics in a running simulation.

In-loop energy modelling. Network simulators such as ns-3 [11] and Castalia [12] include energy consumption models for wireless sensor nodes, but at scales and with assumptions (radio transceivers, mJ-range batteries) that are incompatible with intra-body nanosensors. To the best of our knowledge, no existing MC simulator (including N3Sim [6], BloodVoyagerS [7], and AcCoRD in its original form [3]) integrates per-node energy bookkeeping as an in-loop constraint. If energy is computed only after the run, a depleted nanosensor can still participate in later simulated reactions. Post-processing approaches therefore cannot capture the feedback between energy depletion and system behaviour: when a node runs out of energy, it stops reacting, affecting both the information it propagates and its neighbours. Only an in-loop energy model can correctly capture this coupling.

IV. METHODOLOGY

The extension adds four linked mechanisms to AcCoRD.¹ First, actors store per-unit energy reservoirs, updated at every time step by passive drain and harvest, and charged an event-based cost whenever a reaction fires. Second, molecules carry ownership identifiers so that reaction costs are attributed to the correct nanosensor unit. Third, energy-enabled reactions check the owning unit's energy before firing and are suppressed if it is insufficient, coupling energy state directly to reaction dynamics. Fourth, biomarker identity sets (BmIDSets) propagate through reaction products, enabling the simulator to track information spread and suppress redundant gossip between nanosensors that already share the same biomarker knowledge. Fig. 1 gives an architectural overview of these mechanisms. The JSON configuration format is extended with new optional fields; existing configurations remain fully backward-compatible.

A. Per-Nanosensor Energy Model

Each AcCoRD actor can be declared energy-enabled with the configuration flag "Is Energy Enabled?": true. Six scalar parameters govern the reservoir:

- E_{init} : initial energy at the start of each repeat ("Energy Initial").
- E_{max} : capacity cap; harvested energy above this is discarded ("Energy Max").
- $\dot{E}_{\text{drain}}$: passive energy drain per global microscopic time step Δt ("Energy Drain Passive").
- $\dot{E}_{\text{harvest}}$: passive energy harvest per step ("Energy Harvest Passive").
- e_{det} : energy cost per detection reaction event.
- e_{comm} : energy cost per gossip/communication reaction event.

The discrete-time energy update for a single nanosensor unit u is:

$$E_u(t + \Delta t) = \min(E_u(t) - \dot{E}_{\text{drain}} + \dot{E}_{\text{harvest}}, E_{\text{max}}), \quad (4)$$

¹Source code available at https://github.com/unaestheticc/accord_energy

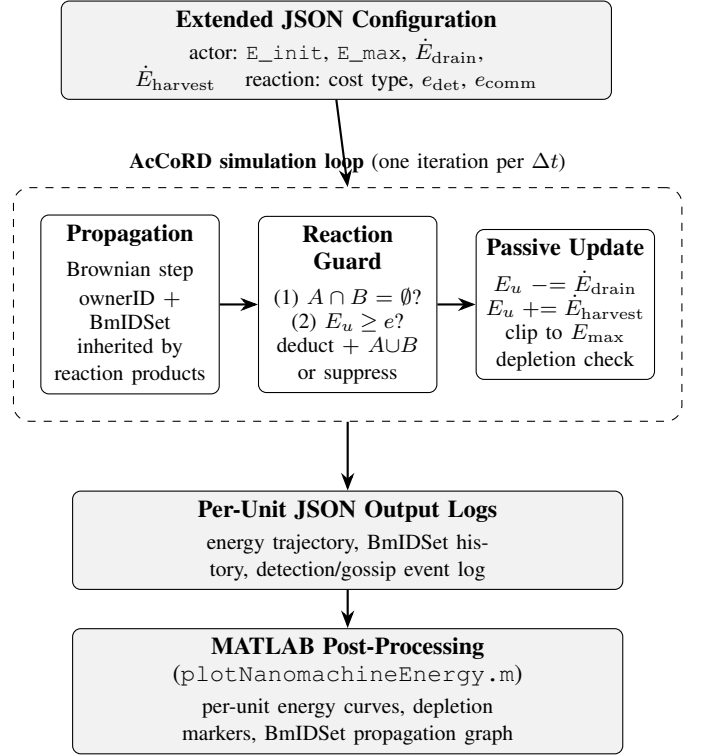


Fig. 1. Architecture of the energy-aware AcCoRD extension.

applied at every global microscopic step, plus instantaneous deductions of e_{det} or e_{comm} whenever a reaction attributable to unit u fires.

A single AcCoRD actor can represent a group of N physically co-located nanosensors. In the BURST modulation scheme, N molecules are emitted at initialisation, each molecule representing one physical nanosensor drifting through the simulation space.

Each actor maintains parallel arrays of length N :

- `energyCurrentPerUnit[u]`: current energy of unit u .
- `bEnergyDepletedPerUnit[u]`: flag set when $E_u < \varepsilon_{\text{min}}$, where ε_{min} is the cost of the least expensive energy-enabled reaction in the simulation (i.e. the gossip cost e_{comm} in the current scenario). This threshold is computed automatically at simulation start-up by scanning all energy-enabled reactions; it is therefore configuration-driven and requires no manual adjustment.

The rationale for this threshold is operational: a nanosensor unit whose remaining energy is strictly below ε_{min} cannot afford even the cheapest available reaction. Marking it as depleted at that point prevents the simulator from futilely attempting and aborting every subsequent reaction involving that unit. Concretely, if the gossip cost is $e_{\text{comm}} = 2 \times 10^{-5}$ pJ, a unit with $E_u < 2 \times 10^{-5}$ is considered depleted. The aggregate `energyCurrent` is the sum of all per-unit levels and is used for display purposes only; all energy decisions operate on individual units.

When a bimolecular reaction involves a molecule owned by unit u of an energy-enabled actor, the cost is charged to

unit u . If a unit is depleted, its molecule cannot participate in energy-gated reactions and is effectively inert. If all N units of an actor are depleted, the actor's next-action time is set to infinity, removing it from the simulation event queue.

To attribute costs to the correct unit, every microscopic molecule carries an `ownerActorID` (16-bit) and `ownerUnitID` (32-bit) identifying the actor and unit that emitted it (-1 if unowned). At burst emission, unit IDs are assigned round-robin ($i \bmod N$) and inherited by reaction products, so every nanosensor-derived molecule retains its owner identity throughout the simulation.

B. Biomarker Identity Tracking (BmIDSet)

A key challenge in nanosensor network simulation is preventing double-counting via gossiping of the same biomarker. To address this, we introduce the `BmIDSet` data type: a compact set of up to 64 globally unique 64-bit biomarker identifiers (IDs) carried by each molecule. Biomarker molecules are assigned a unique ID at creation via an atomic counter (`assignNewBMID()`). Nanosensor molecules begin with an empty `BmIDSet`.

When a detection or gossip reaction fires between molecules with `BmIDSets` A and B , the product molecule receives $A \cup B$ (the set union). Each nanosensor molecule's `BmIDSet` thus monotonically accumulates the identities of all biomarkers encountered, directly or via gossip. The simulator logs this set at each reaction event, enabling reconstruction of the information-spread graph.

Two nanomachines A and B may only have a gossip reaction only when their `BmIDSets` have no overlap: ($A \cap B = \emptyset$, a guard implemented in the microscopic reaction checker), preventing back and forth gossiping reaction.

Reactions are coupled to the energy state through four new fields added to the `AcCoRD` reaction structure:

- `bEnergyEnabled`: whether this reaction incurs energy costs.
- `energyCostType`: string, "detection" or "communication".
- `energyCostValue`: scalar cost per event.
- `energyCostUnit`: "per_reaction" (flat cost) or "per_product" (cost scales with number of products).

Before a candidate reaction fires, two guards run in sequence. First, a `BmIDSet` overlap check verifies that the reacting molecules carry disjoint `BmIDSets` ($A \cap B = \emptyset$): if they already share at least one biomarker identity, the gossip reaction is suppressed because no new information would be exchanged. Second, `canAndConsumeReactionEnergy()` checks that the owning unit has sufficient energy ($E_u \geq e$); if so, the cost is atomically deducted and the `BmIDSet` union $A \cup B$ is assigned to the product; otherwise the reaction is suppressed. This two-stage *reaction guard* couples both information state and energy state directly to reaction dynamics. In the microscopic regime, the unit charged is always the one whose molecule physically participated in the reaction, identified by the `ownerUnitID` field.

For every reaction that fires, the simulator logs per unit: time, current `BmIDSet`, reaction direction (`send/receive/both`), and partner ID. Duplicate events within the same step are suppressed. Per-unit logs are serialised to JSON alongside aggregated statistics (mean/final energy, depletion flag, reaction counts). A companion MATLAB script, `plotNanomachineEnergy.m`, parses this output and produces one subplot per actor with energy curves over time, depleted units highlighted in red, and a dashed threshold line at $\varepsilon_{\min}$.

C. Simulation Scenario

To validate the energy-aware extension, we simulate an intra-body scenario designed to produce heterogeneous reaction activity across nanosensor clusters: a geometry where some clusters are exposed to high biomarker flux and others are largely shielded, so that differential energy depletion is observable within a short simulation window. The scenario models a branching arteriole composed of five rectangular microscopic regions. The arteriole splits at one junction and reconverges at another, creating two parallel flow paths. Clusters positioned near the inlet encounter the full biomarker stream, while downstream clusters receive progressively fewer biomarkers. All regions are simulated in the microscopic regime with a uniform subvolume base size of $1 \mu\text{m}$.

Biomarkers (molecule type 0) are advected at $1.5 \times 10^{-3} \text{ m/s}$ and diffuse with coefficient $D_{\text{BM}} = 2.26 \times 10^{-12} \text{ m}^2/\text{s}$. A total of 56 biomarker molecules are released at $t = 0$ from point sources distributed along the inlet face. Nanosensors (molecule type 1) are placed as thin sheets on the vascular wall in four clusters (actors 56–59), each emitting 25 individual nanosensor molecules at $t = 0$.

Four molecule types model the nanosensor state machine:

- 1) BM ($D = 2.26 \times 10^{-12} \text{ m}^2/\text{s}$, advected): biomarker.
- 2) NM ($D = 5.66 \times 10^{-14} \text{ m}^2/\text{s}$, advected): naive nanosensor.
- 3) NM_{det} (same diffusivity): nanosensor that has detected a BM.
- 4) NM_{gossip} (same diffusivity): nanosensor informed via gossip.

Eight bimolecular reactions implement all state transitions. The three detection reactions (e.g., $\text{BM} + \text{NM} \rightarrow \text{NM}_{\text{det}}$) use binding radius $r_b = 1.025 \mu\text{m}$; the five gossip reactions use $r_g = 2.0 \mu\text{m}$. Binding rates are set to $k = 10^{9999}$ (effectively instantaneous upon contact). All simulation parameters are summarised in Table I.

Note on energy values. The energy constants (E_{init} , $\dot{E}_{\text{drain}}$, e_{det} , e_{comm}) are indicative order-of-magnitude estimates chosen to produce observable depletion within the simulation window. To our knowledge, published biophysical measurements of per-event nanosensor detection and gossip communication energy remain unavailable in the literature; these estimates serve as a physically plausible starting point pending such measurements. The depletion threshold $\varepsilon_{\min} = e_{\text{comm}} = 2 \times 10^{-5} \text{ pJ}$ is derived automatically from the configuration and will update whenever the reaction costs are changed. Calibrating these parameters to real nanosensor hardware is left as future work.

TABLE I
SIMULATION PARAMETERS. ENERGY VALUES (STARRED) ARE ARBITRARY; THE pJ UNITS BROADLY REFLECT THE EXPECTED ORDER OF MAGNITUDE.

Parameter	Value	Unit
Global time step Δt	10^{-5}	s
Simulation duration T	0.04	s
Number of regions	5	—
Number of molecule types	4	—
Diffusion coeff. BM (D_{BM})	2.26×10^{-12}	m ² /s
Diffusion coeff. NM (D_{NM})	5.66×10^{-14}	m ² /s
BM advection velocity	1.5×10^{-3}	m/s
NM advection velocity	0.5×10^{-3}	m/s
Number of biomarker sources	56	—
Nanosensor actors	4	—
Nanosensors per actor (N)	25	—
Total nanosensors	100	—
Binding radius, detection r_b	1.025	μm
Binding radius, gossip r_g	2.0	μm
Initial/max energy E_{init}^*	3×10^{-4}	pJ
Passive drain $\dot{E}_{\text{drain}}^*$	5×10^{-9}	pJ/step
Passive harvest $\dot{E}_{\text{harvest}}^*$	0	pJ/step
Detection cost e_{det}^*	9×10^{-5}	pJ
Gossip cost e_{comm}^*	2×10^{-5}	pJ
Depletion threshold $\varepsilon_{\text{min}}^*$	2×10^{-5}	pJ

V. RESULTS

A. Energy Depletion

Table II summarises the depletion statistics at the end of the 40 ms simulation. Out of 100 nanosensors across four actors, twelve were depleted. No actor lost all its units, meaning that each cluster retained some detection and gossip capability throughout the simulation.

Actor 56 had the highest depletion rate (6 out of 25 units, 24%). This actor is placed near the inlet of the arteriole, where biomarker concentration is highest and reaction rates are therefore greatest. Furthermore, once a biomarker reacts with a nanosensor it is consumed, so fewer biomarkers reach downstream clusters. Actor 57, placed downstream, had a depletion rate of 4 out of 25 (16%). Actors 58 and 59, located furthest from the inlet along the two parallel flow paths, each lost only 1 nanosensor (4%). The position of each actor cluster relative to the biomarker flow thus has a direct impact on energy lifetime.

TABLE II
NANOSENSOR ENERGY DEPLETION SUMMARY (SINGLE RUN, SEED = 5)

Actor ID	NMs	Depleted	Depl. rate
56	25	6	24%
57	25	4	16%
58	25	1	4%
59	25	1	4%

Table III lists four representative nanosensors from actor 56. The correlation between reaction activity and energy depletion is clear. NM 2, which participated in no reactions, retained a final energy of 2.80×10^{-4} pJ, close to its initial value (the

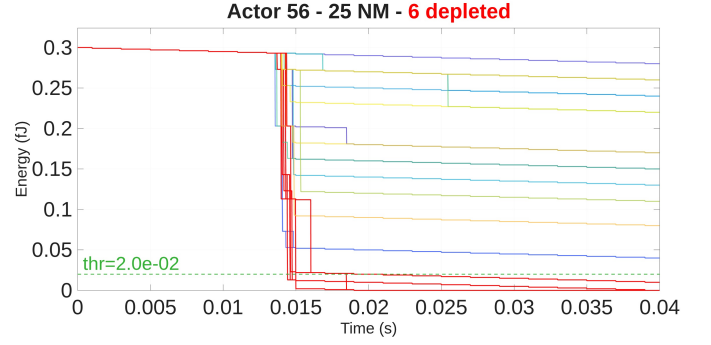


Fig. 2. Energy trajectories of 25 nanosensors of Actor 56 (seed = 5). Red curves indicate depleted units; the green dashed line marks the depletion threshold $\varepsilon_{\text{min}} = 2 \times 10^{-5}$ pJ

small deficit is due to passive drain). NM 22 (1 detection, 1 gossip) was not depleted. NM 11 (2 detections, 5 gossips) and NM 7 (3 detections, 1 gossip) were both fully depleted, illustrating that depletion can result from either many low-cost gossip events or fewer high-cost detection events. Because the energy values are indicative order-of-magnitude estimates, no attempt is made to reconcile the exact numerical differences with a physical model; what matters is that the qualitative ordering is correct: more active nanosensors lose energy faster, and the simulation enforces this in-loop so that depleted nanosensors can no longer react.

TABLE III
REPRESENTATIVE NANOSENSORS OF ACTOR 56 (SEED = 5)

NM	Detections	Gossips	E_{final} (pJ)	Depleted?
NM 2	0	0	2.80×10^{-4}	No
NM 22	1	1	1.70×10^{-4}	No
NM 11	2	5	0	Yes
NM 7	3	1	0	Yes

B. BmIDSet Propagation

Table IV traces the BmIDSet evolution of NM 6 (actor 56): starting from an empty set, it first received BM#26 via gossip, then directly detected BM#34, #22 and #33 across two events, propagating its accumulated knowledge to neighbours after each step before becoming depleted. The absence of any repeated biomarker ID in any unit's log confirms the deduplication property of the BmIDSet union mechanism.

C. Discussion

Energy parameter scale. The energy values used in this experiment are indicative order-of-magnitude estimates in picojoules, chosen to produce observable depletion within the simulation window and to verify that the energy-gating logic operates correctly. The pJ range is consistent with the scale of bio-electrochemical processes at the nanoscale; to our knowledge, validated measurements of per-event nanosensor detection and gossip communication energy remain to be reported in the literature. The AcCoRD energy extension is

TABLE IV
BMIDSET EVOLUTION OF NM 6 (ACTOR 56, SEED = 5)

Time (ms)	Event	Partner	BmIDSet
13.71	gossip ←	NM 14	{26}
14.16	detect	BM#34, #22	{26, 34, 22}
14.17	gossip →	NM 18	{26, 34, 22}
14.18	gossip →	NM 14	{26, 34, 22}
14.41	detect	BM#33	{26, 34, 22, 33}
14.42	gossip →	NM 14	{26, 34, 22, 33}
14.43	gossip →	NM 18	{26, 34, 22, 33}

designed to accommodate any value as such measurements become available.

Binding radius and molecular geometry. The two binding radii used in the scenario reflect the physical sizes of the molecules involved in each reaction type. In AcCoRD’s microscopic regime, the binding radius r_b represents the centre-to-centre distance below which two molecules are considered to have reacted; for a bimolecular reaction between species of radii r_1 and r_2 , the natural choice is $r_b \approx r_1 + r_2$. For the detection reactions (BM+NM), the biomarker is small compared to the nanosensor, yielding $r_b = 1.025 \mu\text{m}$. For the gossip reactions (NM+NM_{det}), both reactants are nanosensor-sized molecules, so their combined radii give a larger binding distance $r_g = 2.0 \mu\text{m}$. This choice ensures that the reaction geometry is consistent with the relative sizes of the interacting species rather than being an arbitrary tuning parameter.

Single gossip per biomarker identity. The BmIDSet overlap guard prevents two nanosensor molecules from triggering a gossip reaction if they already share at least one common biomarker ID. This ensures that once NM A has transmitted knowledge of biomarker b to NM B , a subsequent physical encounter between the same pair does not produce a redundant and energy-consuming re-transmission of information that NM B already holds.

This is a deliberate modelling choice, but it may not reflect the physical reality of molecular communication. In MC, nanosensor molecules diffuse freely and can repeatedly come within binding radius of one another; without an explicit molecular memory (e.g. a conformational change or covalent modification that marks the molecule as “already informed”), there is no biophysical mechanism to suppress re-reaction. A more conservative model would allow every encounter to trigger a gossip reaction regardless of prior history, which would increase energy consumption without adding new information to the network. Whether the BmIDSet guard is a reasonable abstraction or an over-idealisation depends on the nanosensor architecture assumed; this remains an open modelling question for future work.

Depletion and information loss. NM 6 (2 detections, 5 gossips) depleted fully and ceased to participate in the reaction system. NM 7 reached the same outcome through a different path: three detections and only one gossip event sufficed to exhaust its reservoir, because each detection costs 9×10^{-5} pJ compared to 2×10^{-5} pJ for gossip. Any subsequent biomarkers reaching these nanosensors’ locations were not detected,

and they no longer propagated gossip to their neighbours. This information-loss effect cannot be reproduced by any post-processing energy model and demonstrates the necessity of an in-loop energy gate.

VI. CONCLUSIONS

We have shown how nanosensor energy can be represented as a per-unit state variable within reaction-diffusion simulation, updated at every time step and used to gate detection and gossip reactions in-loop. This coupling ensures that a depleted nanosensor immediately stops participating in the reaction system, an effect that post-processing approaches cannot replicate. The approach is implemented in AcCoRD through per-unit energy reservoirs, molecule ownership tracking, energy-gated reactions, and a BmIDSet mechanism for information deduplication, with a backward-compatible JSON interface.

Simulation of a branching arteriole scenario with 100 nanosensors and 56 biomarkers confirms the expected behaviour: nanosensors with higher reaction activity deplete faster, the BmIDSet mechanism faithfully tracks information propagation, and energy depletion silently removes exhausted nodes from the system dynamics. The results demonstrate that in-loop energy modelling changes the simulation outcome in ways that cannot be recovered after the fact. The implementation is publicly available at https://github.com/unaestheticc/accord_energy; a README covers the build process, JSON configuration, and scenario reproduction.

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